



Possible ionospheric anomalies associated with the 2009 M_w 6.4 Taiwan earthquake from DEMETER and GNSS TEC

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Abstract

The recent advances in satellite based ionosphere monitoring provided more evidence about earthquake (EQ) precursors from lithosphere ionosphere coupling mechanism. In this paper, we investigate ionospheric anomalies associated with the December 19, 2009, Taiwan EQ ($M_w=6.4$), that occurred on UT = 13:02 h. Seismo ionospheric anomalies are evaluated during 31 days (20 day before followed by 10 days after the EQ) in the data of French Satellite Detection of Electro-Magnetic Emissions Transmitted from Earthquake Regions (DEMETER) and validated from Global Navigation Satellite System (GNSS) retrieved Total Electron Content (TEC). The day time values of ISL (Langmuir Probe) and IAP (Ion Analyzer) from DEMETER point to significant ionospheric perturbations within 5 days associated with EQ day. Moreover, GNSS stations of International GNSS Service (IGS) within EQ preparation zone also validate anomalies in ionosphere prior/after the M_w 6.4, Taiwan EQ. Moreover, spatial and temporal analyses of daily ionospheric indices from DEMETER and GNSS point to simultaneous enhancement in electron density, ion density, electron temperature and TEC related to the main shock during quiet geomagnetic storm ($Dst < -30$ nT, $K_p \leq 3$). All these evidence aids in promoting the lithosphere- ionosphere coupling over the epicenter of M_w 6.4 during the preparation period from DEMETER and GNSS TEC.

Keywords DEMETER · GNSS · TEC · Earthquake

1 Introduction

There are many reports about EQ induced anomalies in ionosphere from different satellites observations and a correlation is established with lithosphere before and after the main shock (Hayakawa and Hobara 2010; Kuo et al. 2014; Guo et al. 2015; Zhang et al. 2017; Contoyiannis and Potirakis 2018; Florios et al. 2020; Ulukavak and Inyurt

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2020; Elshin and Tronin 2020). The rapid advancement in satellite based EQ precursors plays a decisive role in determining different atmospheric and ionospheric perturbations with different GNSS measurements (local and global) and other remote sensing procedures (Afraimovich et al. 2004; Tariq et al. 2019; Marchetti et al. 2019; Sekertekin et al. 2020). Moreover, many reports also manifested EQ perturbations over the epicenter from DEMETER, which is dedicated to monitor seismo ionospheric anomalies (Ryu et al. 2014; Tao et al. 2017; Shah et al. 2019).

On the other hand, synchronized and collocated ionospheric anomalies over the epicenter of future EQ from TEC and other satellites are measured in different statistical analysis (Parrot 2011; Shah and Jin 2015; Ke et al. 2016; Liu and Xu 2017; Daneshvar and Freund 2017). Similarly, different morphological characteristics and its propagation to ionosphere subsequent to the EQ are studied for seismo ionospheric anomalies in different case studies (Shah and Jin 2018; Ahmad et al. 2019; Shah et al. 2020b). For example, Akhoondzadeh et al. (2010) implemented different mathematical and statistical procedures on multiple ionospheric data to designate significant perturbations (positive and negative anomalies) over the EQ preparation zone within few days before and after the main shock. Ryu et al. (2014) formulated DEMETER's data in the form of differential measurement between equatorial and low latitude ionosphere and further showed enhancement in that differential ionospheric values related to EQs. Similarly, anomalous pattern in the form of enhancement and depletion in DEMETER's measurements are also investigated for ionospheric precursors related to EQ in different regions (Parrot et al. 2006; Sarkar et al. 2007; Zhang et al. 2009). Parrot (2011) statistically analyzed the perturbations in ion density in DEMETER's satellite measurements with different global seismic activities. Some reports strongly correlated ionospheric anomalies with geomagnetic storm (Dautermann et al. 2007; Afraimovich et al. 2008; Liu et al. 2008; Thomas et al. 2016) and some even rejected the phenomena of EQ's ionospheric anomalies (e.g., Geller et al. 1997; Rishbeth 2007).

Ionospheric anomalies over EQ epicenter can be discussed in two different ways: Positive Holes (p hole) emission (Freund et al. 2007, 2009) and Radon emanation from EQ prone regions (Pulinets et al. 2006; Pulinets and Ouzounov 2011). According to Freund et al. (2007), stressed rocks beneath the Earth surface due to tectonic force trigger p holes emission to the surface, which lead to air ionization near the lithosphere. Subsequently, this phenomenon raises up to the atmosphere and ionosphere at different altitudes to cause deviation in radio signal propagation (e.g., GPS and other satellite delays). On the other hand, Radon emanation from EQ epicenter and associated fault lineaments due to future main shock preparation period also caused air ionization, ion recombination and subsequently propagate to ionosphere as a result of seismo ionospheric anomalies (Pulinets et al. 2006; Pulinets and Ouzounov 2011). As the amount of Radon reached the threshold, it disrupts radio signals during the coupling of lithosphere atmosphere ionosphere over the EQ epicenter.

In this study, we investigate ionospheric anomalies before and after the Taiwan EQ of M_w 6.4 from DEMETER and evidence is provided by GNSS TEC to manifest a stringent proof to seismo ionospheric anomaly. Taiwan Island has the high occurrence rate of EQ in East Asian region, which is relatively higher than any other region of this territory (Liu et al. 2006). We investigate seismo ionospheric anomalies during quiet geomagnetic storm conditions ($Dst < -30$ nT, $K_p \leq 3$), which aim to show the bona fide seismic anomalies in the data of DEMETER and GNSS TEC. This paper is structured as: Sect. 2 includes precise data and method of monitoring seismo ionospheric anomalies. Sections 3 and 4 comprise of results and discussions and conclusions of all the findings, respectively.

2 Data and method

In this paper, we investigate ionospheric anomalies associated with M_w 6.4 EQ from DEMETER and TEC of different GNSS stations in the seismogenic region of the impending EQ (Fig. 1). We retrieve EQ information from the United States Geological Survey via website (<https://earthquake.usgs.gov/earthquakes/search>). Moreover, geomagnetic storm indices are retrieved from International Services of Geomagnetic Indices (ISGI) (<https://isgi.unistra.fr/index.php>) to differentiate seismic and storm anomalies during December, 2009 (Fig. 2). In this paper, we study ionospheric anomalies from the DEMETER's flights within the EQ preparation zone over the epicenter of the Taiwan EQ. The preparation zone of this EQ is estimated by Dobrovolsky et al. (1979) formula, as below.

$$R = 10^{0.43M_w} \text{ km} \quad (1)$$

where M_w and R are EQ magnitude and radius of epicentral zone, respectively ($R = 565$ km for this EQ). From the above equation, it is clear that seismogenic zone is dependent on EQ magnitude. For example, EQ's large magnitude retains large critical zone around the epicenter and vice versa (Shah et al. 2020c). DEMETER's ISL and IAP data over the epicenter of impending EQ are analyzed for prior the main shock and after the EQ. Moreover, seismic anomalies in ionosphere are investigated from those DEMETER's flight, which occur within the Dobrovolsky et al. (1979) region.

DEMETER is a French satellite, launched in 2004 by French Space Agency (CNES) for the monitoring of EQ induced magnetic and electrical disturbances in ionosphere. It has several magnetic and electrical sensors, energetic and particle analyzer, ion spectrometer and Langmuir probe (Cussac et al. 2006). Moreover, its orbits cover the Earth's latitude between 65° S and 65° N in 32 flights (16 ascending and 16 descending) (Parrot et al. 2005). In this paper, we obtained daily temporal data in 1 s resolution format from ISL

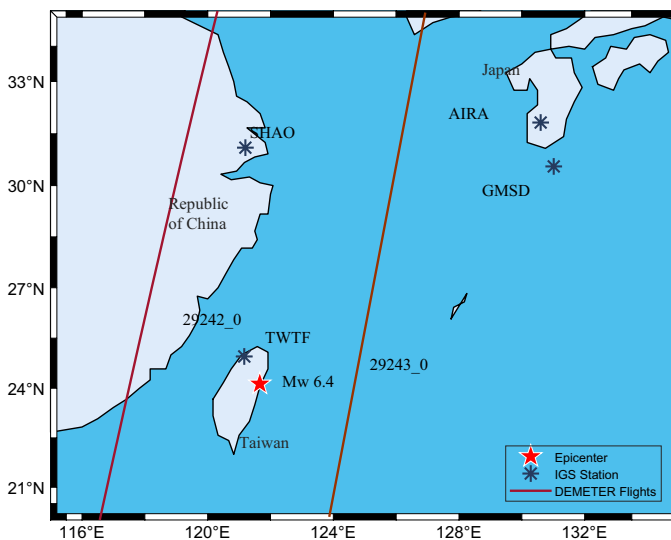


Fig. 1 Geographic location of the EQ epicenter by a red star, while the black asterisks are IGS ground stations and two lines are DEMETER flights with the orbit number on EQ day

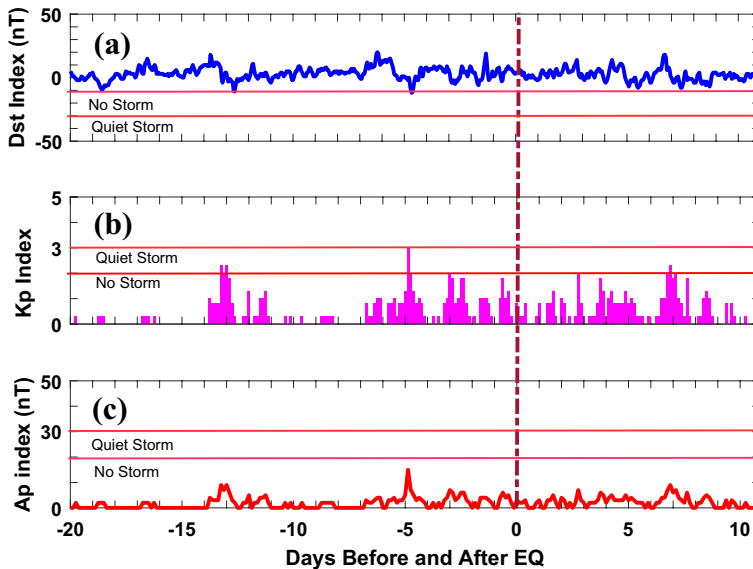


Fig. 2 Geomagnetic storm conditions from Dst, Kp and Ap indices before and after the EQ day. Red dashed line represents EQ day

and IAP onboard DEMETER satellite for detecting seismo ionospheric anomalies. ISL provides the main parameters like, electron density, ion density and electron temperature (Lebreton et al. 2006). However, IAP has O^+ , H^+ and ion temperature data for sensing EQ induced variations in the ionosphere. In this study, different ionospheric indices for 20 days and 10 days after the main shock are extracted from DEMETER's ISL and IAP to detect seismo ionospheric anomaly. More information and data extraction of DEMETER can be accessed through (<https://cdpp-archive.cnes.fr>).

The daily values of DEMETER are processed between the confidence bounds of median and standard deviation to identify the EQ anomaly. These bounds are obtained from the Median (M) and 2σ (σ =standard deviation) of all the days for ISL and IAP of DEMETER, as in Sarkar et al. (2012). The bounds are obtained from all the values in this study before and after the EQ main shock (i.e., 20 days before and 10 days after), as below;

$$\text{Upper Limit} = M + 2\sigma \quad (2)$$

$$\text{Lower Limit} = M - 2\sigma \quad (3)$$

where M is median value and σ is the standard deviation of the data. The reason of adding this bounds technique is no uniform diurnal data from DEMETER like GNSS TEC. To validate DEMETER results, we estimate TEC anomalies for the same EQ from 4 GNSS stations of IGS network operating within the seismogenic zone. The temporal and spatial TEC anomalies are analyzed beyond the confidence bounds for the Taiwan EQ during December, 2009. Receiver Independent Exchange Format (RINEX) files from the 4 different IGS stations (i.e., AIRA, MSD, SHAO and TWFT) are obtained from <ftp://cddis.gsfc.nasa.gov/pub/gps/products/ionex>. Slant TEC from the GNSS ground stations are obtained along the ray path from satellite to receiver (Li et al. 2014; Ciraolo et al. 2007). In order to examine

VTEC, Slant Total Electron Content (STEC) has been extracted from IGS ground stations from the equation.

$$\text{STEC}_a^h = -\frac{f_1^2 f_2^2}{40.3(f_1^2 - f_2^2)} \left(P_{4,a}^h - c \times \text{DCB}_a - c \times \text{DCB}^h \right) \quad (4)$$

From Eq. 4, f_1 and f_2 are the GPS dual frequencies, c is speed of light, $P_{4,a}^h$ is the differences between smoothed coded measurements, while DCB_a and DCB^h are differential code biases for the satellite and receiver, respectively (Sekertekin et al. 2020). Furthermore, to convert STEC into VTEC, the following equation has been applied to STEC:

$$\text{VTEC} = \text{STEC} \times \cos \left(\arcsin \left(\frac{R \sin z}{R + H} \right) \right) \quad (5)$$

In this equation, H is the height of the top of ionospheric layer and R is the radius of Earth and z is the zenith angle of the satellite (Klobuchar 1987) and it is represented in TEC unit (where, $1 \text{ TECu} = 10^{16} \text{ el/m}^2$).

The VTEC from every station within the seismogenic zone is retrieved by Cirao et al. (2007) method and seismic anomalies are studied in the diurnal observations for 15 days before and 5 days after the main shock (Fig. 5). To detect abnormalities in VTEC, confidence bounds are obtained for every day before and after the main shock. For every day of GNSS TEC, bounds are acquired from the median, upper and lower quartile of 20 days (10 days before and 10 days after the observed day), as in Liu et al. (2009). The identified confidence bounds are showed in the following equations:

$$\text{Upper Bound} = M + 1.5(\text{UQ} - M) \quad (6)$$

$$\text{Lower Bound} = M - 1.5(M - \text{LQ}) \quad (7)$$

where M is the Median, UQ is upper-quartile range and LQ is lower-quartile of the normal TEC distribution. Deviation in TEC (dTEC) is used to further intensify the TEC variations, as in Tariq et al. (2019) by the following equations:

$$\text{dTEC} = \begin{cases} \text{TEC} - X_{\text{LB}}; & \text{TEC} < X_{\text{LB}} \\ 0; & X_{\text{LB}} \leq \text{TEC} \leq X_{\text{UB}} \\ \text{TEC} - X_{\text{UB}}; & \text{TEC} > X_{\text{UB}} \end{cases} \quad (8)$$

In above equation, X_{UB} and X_{LB} are upper and lower bounds, respectively. As per previous findings (Shah et al. 2020a), temporal DEMETER and TEC beyond upper confidence bound is termed as positive ionospheric anomaly and daily values below lower bound is termed as negative anomaly. The deviations on each anomalous day are summarized in Tables 1 and 2.

3 Results and discussion

In this study, ionospheric anomalies are investigated from DEMETER satellite and GNSS TEC associated with the Hualien EQ in Taiwan on December 19, 2009. Anomalies observed in the ionosphere over the seismogenic zone of impending main shock are due to

Table 1 TEC anomalies information of the 2009 M_w 6.4 Hualien EQ

S. no	GNSS stations	Distance from epicenter (km)	Geographic coordinates		Anomalous days within EQ period and % deviation			
			Latitude (°)	Longitude (°)	Pre-seismic day	%Deviation	Post seismic day	%Deviation
1	Aira	1233	31.8	130.5	0	4	Nil	–
2	Grnsd	1174	30.5	131.1	0	4	Nil	–
3	Shao	830	31.1	121.2	–9	2	Nil	–
4	Twft	117	24.9	121.3	–9, 0	3, 5	+3, +4	3, 4

Table 2 Anomalies from different DEMETER instruments in this study

S. no	DEMETER sensors	Parameters of different Sensors	Anomalous days within EQ period	
			Pre-seismic	Post-seismic
1	ISL	Electron density	– 1, 0	+ 2
		Electron temperature	– 1	Nil
		Ion density	– 1, 0	+ 2, + 5
2	IAP	H ⁺ density	Nil	+ 1
		O ⁺ density	– 1	+ 1, + 2
		Ion temperature	Nil	Nil

lithosphere ionosphere coupling. Nevertheless, the temporal and spatial analysis of GNSS TEC and DEMETER manifest sufficient evidence of a series of ionospheric anomalies within 5–10 days prior and after the EQ. These anomalies correspond to the main shock in VTEC, dTEC, ISL and IAP in the measurements of GNSS and DEMETER. The use of a lead time of around 20 days is consistent with the finding of Yang et al. (2020) that they could detect criticality one to two weeks before the 2016 Kumamoto EQ in Japan by studying criticality in GNSS surface deformation as a possible agent for exciting atmospheric gravity waves (AGW) in the atmosphere and GNSS fluctuations in the frequency range of AGWs with the use of natural time analysis which is a new method to identify criticality either in EQ precursory phenomena (Varotsos et al. 2011).

Figures 3 and 4 shows different parameters of DEMETER's ISL and IAP with confidence bounds for the investigation of EQ anomalies for 31 days (20 day before the EQ day and then 10 day after. In Figs. 3 and 4, DEMETER data for EQ year of December 2009 and two other years (2007 and 2008 of no EQ) are showed to differentiate the variations of seismic event from non-seismic activity. From ISL on board DEMETER satellite, significant ionospheric perturbations beyond the bounds within 5 days before and after the main shock are clear in the analysis of electron density and ion density, which may be triggered by the EQ. Moreover, H⁺ density and O⁺ density also show synchronized anomalies to ISL data consequent to the impending main shock (Fig. 4). To quantify the variation in ISL and IAP from the normal distribution, percentage deviation show more than 10% enhancement over the bounds. The anomalies are differentiated from the rest of normal distribution by the implementation of bounds method of median and standard deviation on the daily values of both ISL and IAP. On the other hand, electron temperature in ISL show low intensity anomaly (Fig. 3c) and ion temperature from IAP has no significant deviation (Fig. 4c). It indicates less visible variation in electron and ion temperature, however, other indices like electron density and ion density show clear and huge anomalies related to the EQ. Shah et al. (2019) also reported anomalous ionospheric behavior of electron density and electron temperature from DEMETER subsequent to the main shock. They also analyzed anomalies on EQ latitude as compared to its conjugate latitude axis from the analysis of different ionospheric indices in DEMETER data. One can see clear day time ionospheric anomalies associated with M_w 6.4 EQ from DEMETER during quiet storm days, which can be the bona fide seismo ionospheric anomalies.

To provide more evidence to the coupling of lithosphere and ionosphere during the preparation period of M_w 6.4 EQ, TEC anomalies are investigated from 4 GNSS stations

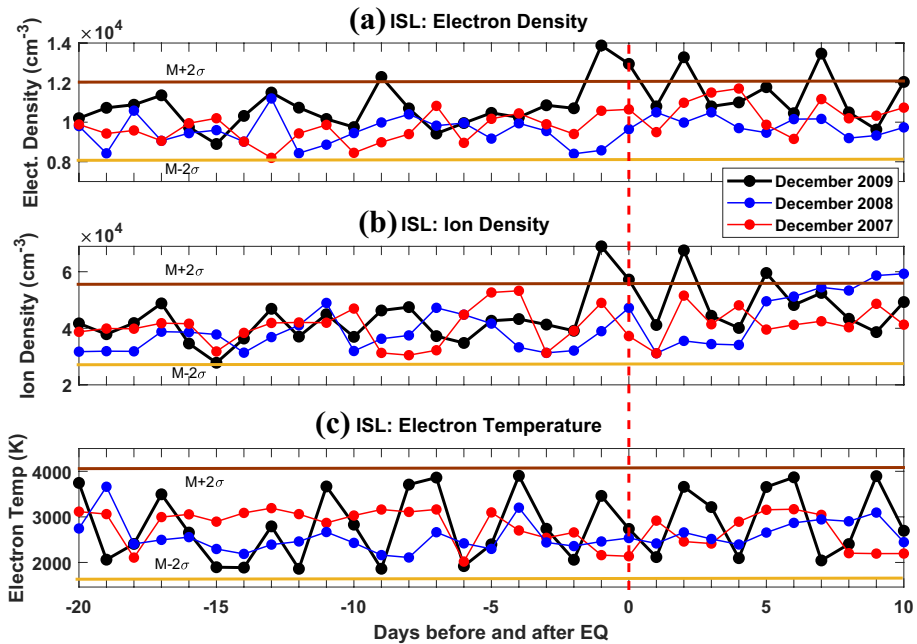


Fig. 3 ISL retrieved electron density, ion density and electron temperature for 20 days before and 10 days after the EQ on December 9, 2009. The same days from December 2007 and 2008 are also showed with no EQs for quantifying ionospheric variations. EQ day is represented by a vertical dashed line

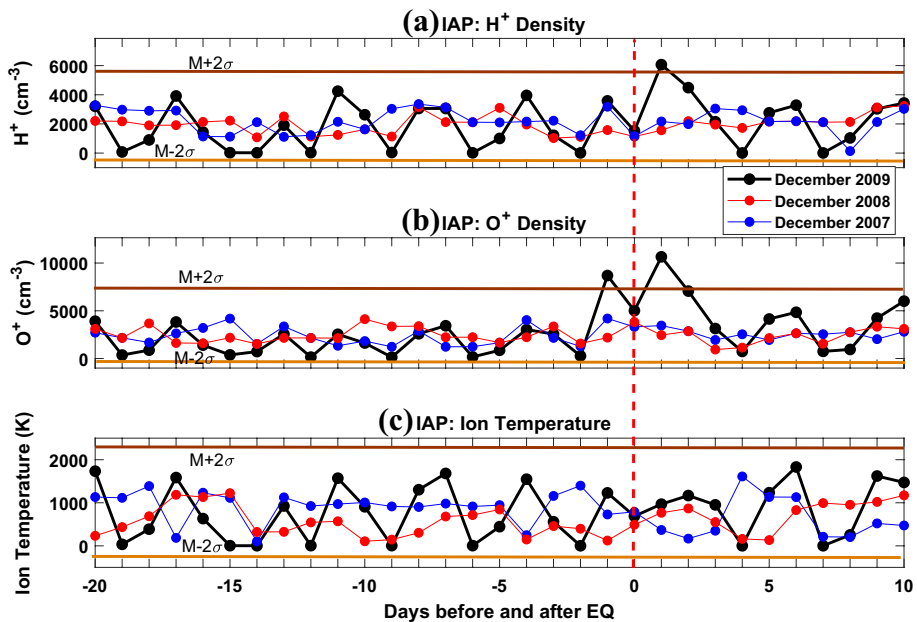


Fig. 4 Hydrogen plus, oxygen plus and ion temperature from IAP on board DEMETER related to the Taiwan EQ. DEMETER data values from December 2007 and 2008 are also showed to discriminate EQ and non-EQ variations. The EQ day is highlighted by a red dashed line

(e.g., AIRA, GMSD, SHAO and TWFT stations) within the critical region of the Taiwan EQ (Figs. 5, 6). Temporal and spatial series show clear and consistent enhancement of TEC anomalies on the main shock day during 5 days before/after the main shock. The ionospheric anomaly on main shock day is clearly visible in all GNSS station, confirms possible seismo ionospheric anomaly. It is observed that the anomalies in TEC occurred during UT = 04–10 h during the EQ preparation period. Moreover, SHAO station shows no anomaly like other stations on main shock day (Fig. 5d). This may be due to no exhalation of energy of the EQ over the ionosphere of SHAO station, as seismic event is a regional phenomenon. Furthermore, spatial TEC in the form of differential TEC also show clear anomalies in three stations in Fig. 6. In Figs. 5 and 6, positive deviations of TEC occur for more than 5 TECU from the rest of normal measurements.

The integrated analysis of DEMETER (ISL and IAP) and GNSS TEC manifest clear variation in the window of 5 days before/after the EQ as a result of preparation process (Fig. 7). All these ionospheric indices endorse the findings of individual analysis for the enhancement related to main shock in DEMETER and TEC. Figure 7a shows intensive variations in all GNSS stations in 5 days before/after the main shock with a high value of > 5 TECU from the normal distribution except Shao station. This deviation may occur due to the main shock energy during the preparation period, as significant variations are also clear in the analysis of DEMETER estimation (Fig. 7b, c). On the other hand, geomagnetic storm indices remain quiet for 20 days prior to EQ, and similarly, 10 days after and it further assists to correlate EQ anomalies with seismic activity during quiet storm days.

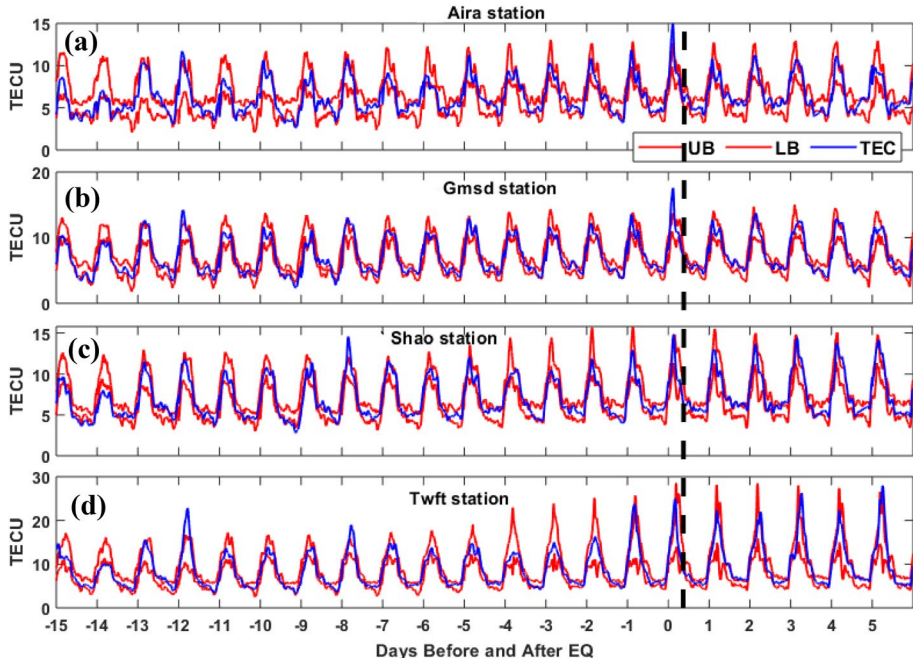


Fig. 5 Four different IGS stations are used to analyze TEC variations. Blue line represents daily TEC values from different stations around the epicenter, while red lines are for upper and lower bounds. EQ day is represented by a vertical dashed line with heading of all GNSS stations

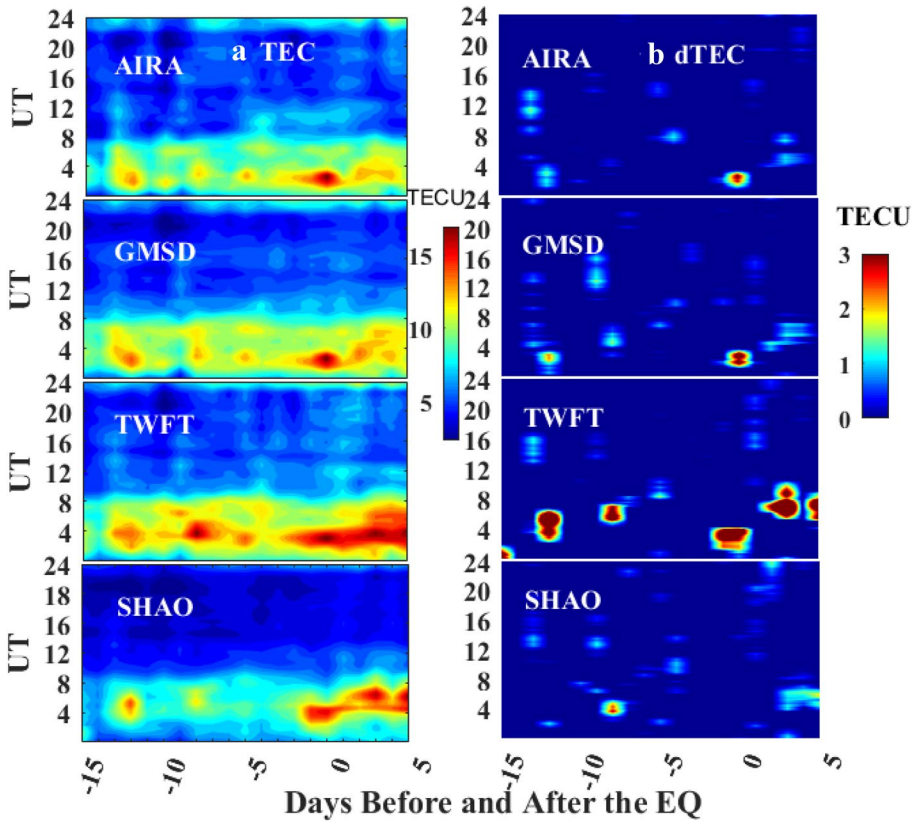


Fig. 6 Spatial VTEC (left) and dTEC (right) have been analyzed within the seismogenic zone for 15 days before and 10 days after the EQ

Our analysis of ionospheric anomalies from multiple indices portray EQ energy in ionosphere for the investigation of lithosphere ionosphere coupling hypothesis. This study also shows significant path of propagation of ionospheric anomalies in the lower and upper ionosphere from TEC and DEMETER satellite. The daily TEC values are obtained at 400 km above the epicenter and DEMETER data is analyzed at ~700 km in the ionosphere. All these lower and upper ionosphere anomalies have source from inside the Earth at seismogenic zone. According to Freund (2010), the more influx of p holes due to squeezed rock over the epicenter could deviate the propagation of radio signal in the ionosphere (Freund 2010). As the main shock day arrives, the more and more p holes emission from seismogenic zones effect air ionization in the lower atmosphere followed to ionosphere. Moreover, Pulnits and Ouzounov (2011) reported the coupling of lithosphere and ionosphere as a result of Radon gas emission from EQ prone regions during the preparation period. Following these hypothesis, Ryu et al. (2014) investigated seismo ionospheric anomalies within few days before the main shock during the seismogenic period from DEMETER and validated by TEC. Similarly, Shah et al. (2019) also reported significant perturbations of ionospheric clouds over EQ latitudinal axis than its conjugate axis from the analysis of DEMETER in the Chile region. Kuo et al. (2011) reported enhancement and depletion in ionospheric anomalies as a result of variations in the electric field around the epicenter by

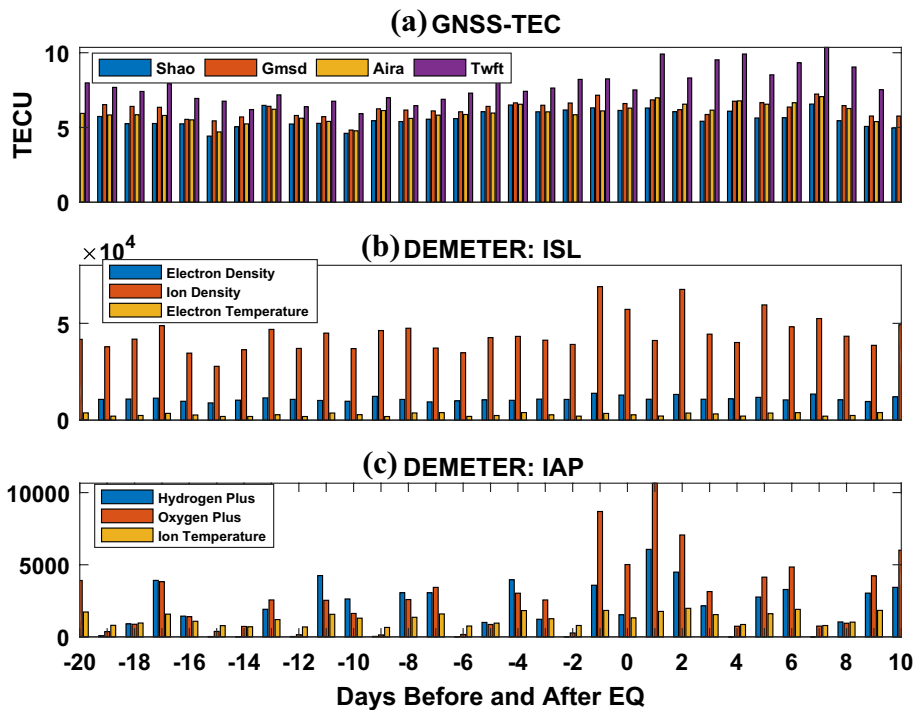


Fig. 7 Comparison of daily VTEC from all GNSS stations and DEMETER satellite for variations consequent to the Taiwanese EQ

changing the direction of flow of current from squeezed rocks. Later on, Kuo et al. (2014) simulated the flow direction of ionospheric anomalies by introducing any arbitrary path of potential difference to improve the lithosphere ionosphere coupling. Furthermore, they suggested more visibility of ionospheric anomalies in the form of positive and negative anomalies due to plasma drift from EQ epicentral region. In this study, DEMETER satellite mission provided sufficient evidences to correlate ionospheric anomalies with EQ as a precursory phenomenon. Although, these results suggest significant correlation of ionospheric anomalies with EQ but still more precursory analyses require to show an influential role in the forecasting of EQs.

4 Conclusions

In this study, seismo ionospheric anomalies are analyzed by utilizing the data of DEMETER (at 700 km altitude) and GNSS TEC (400 km altitude) from the IGS network before and after the M_w 6.4 EQ in the Taiwan region. Statistical bound method is implemented on the daily values of different ionospheric indices to differentiate seismic precursors from storm induced anomalies. Results shows significant evidence of the coupling of lithosphere and ionosphere prior and followed by the main shock during the preparation period. Following are the main outcomes from this study:

- ISL sensor of DEMETER manifests abnormal electron and ion density within 5 days before and after the main shock consequent to the EQ. Whereas, electron temperature shows no clear anomaly and remains within the bounds during the estimated period. It means that EQ varied ionospheric electron and ion due to enormous energy exhalation during the main shock preparation period.
- In case of IAP sensor, day time H^+ and O^+ densities also show significant perturbations before and after the main shock and ion temperature shows no abnormality before and after the impending EQ.
- GNSS TEC also shows perturbations in spatial and temporal analyses during UT=04–10 h on the main shock day. Only, SHAO station shows no anomaly like other stations on the main shock day. The variations in GNSS TEC validate the precursory enhancements of DEMETER within the EQ preparation period. The outcomes of dTEC show that positive anomalies are more dominant than negative anomalies consequent to the EQ. This study investigates ionospheric anomalies during quiet geomagnetic conditions. However, more work needs to investigate the ionospheric anomalies with multiple ground and space based data to delineate the phenomena.

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Compliance with ethical standards

Conflict of interest The authors have no conflicts with their organizations and any other funding agency.

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